

File Organization and Database Systems

Number of Credits: 3 (3-0-0)

Total Number of Hours: 40

Prerequisites: Programming Languages

Introduction: Purpose of database systems, data abstraction and modelling, instances and schemes, database manager, database users and their interactions, data definition and manipulation language, data dictionary, overall system structure. [3 Lectures]

Entity-relationship model: Entities and entity sets, relationships and relationship set, mapping constraints, E-R diagram, primary keys, strong and weak entities, reducing E-R diagrams to tables, trees or graphs, generalization and specialization, aggregation. [5 Lectures]

Brief Introduction to hierarchical and network model: Data description and tree structure diagram for hierarchical model, retrieval and update facilities, limitations; Database task group (DBTG) model, record and set constructs retrieval and update facilities, limitations. [6 Lectures]

Relational model: Structure of a relational database, operation on relations, relational algebra, tuple and domain relational calculus, salient feature of a query language. Structured query language: Description an actual RDBMS and SQL. [4 Lectures]

Normalization: Pitfalls in RDBMS, importance of normalization, functional, multivalued and join dependencies, INF to 5NF, limitations of RDBMS. [4 Lectures]

Database tuning: Index selection and clustering, tuning of conceptual schema, denormalization, tuning queries and views. [4 Lectures]

Query optimization: Importance of query processing, equivalence of queries, cost estimation for processing a query, general strategies, birelational and multi-relational join algorithms, algebraic manipulation. [6 Lectures]

Crash recovery: Failure classification, transactions, log maintenance, check point implementation, shadow paging, example of an actual implementation. [5 Lectures]

Concurrency Control in RDBMS: Testing for serializability, lock based and timestamp-based protocols; Deadlock detection and Recovery. [3 Lectures]

